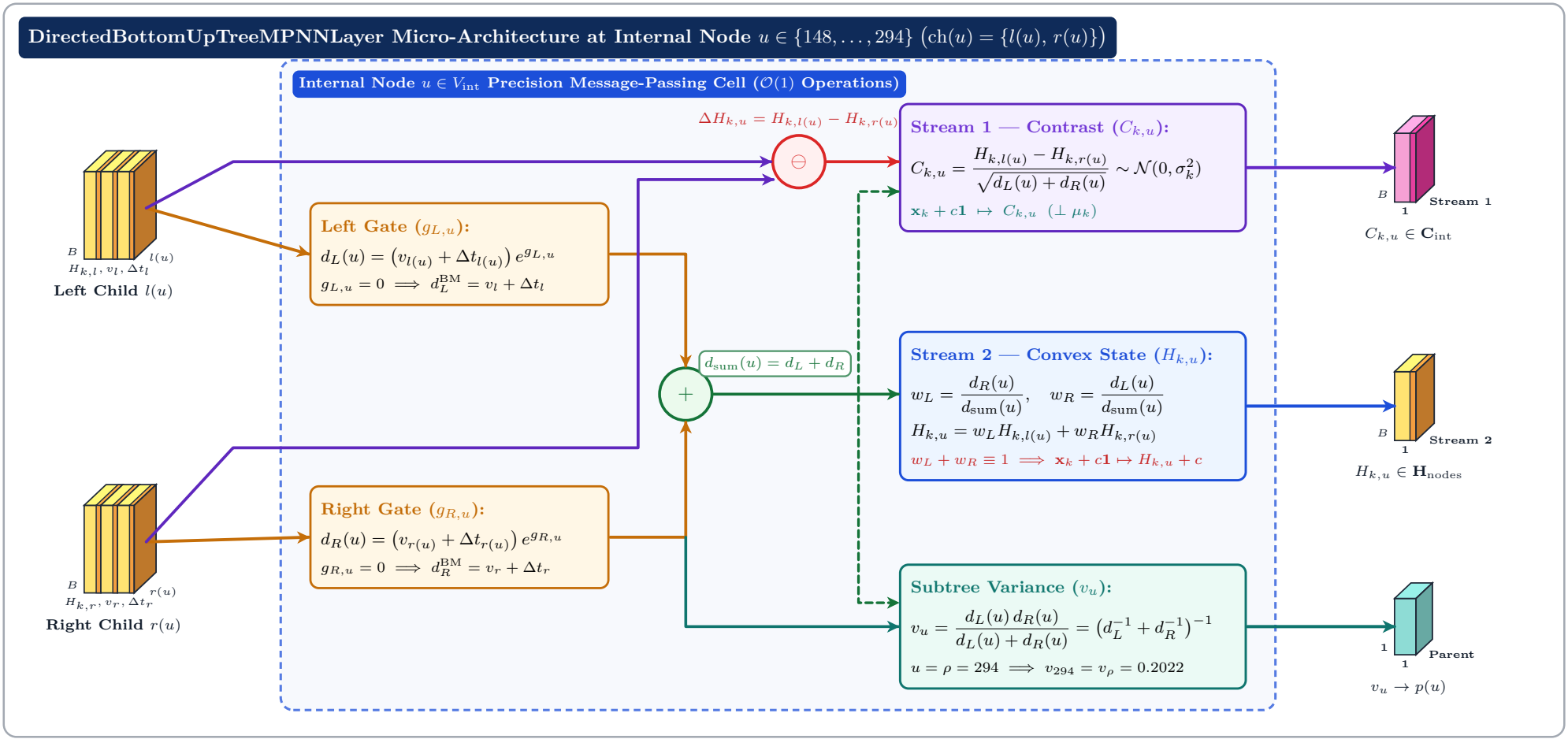

DirectedBottomUpTreeMPNNLayer Micro-Architecture at Internal Node $u \in \{148, \ldots, 294\}$ $(\mathrm{ch}(u) = \{l(u), r(u)\})$
Internal Node $u \in V_{\mathrm{int}}$ Precision Message-Passing Cell ($\mathcal{O}(1)$ Operations)
$\Delta H_{k,u} = H_{k,l(u)} - H_{k,r(u)}$
Stream 1 — Contrast ($C_{k,u}$):
$C_{k,u} = \frac{H_{k,l(u)} - H_{k,r(u)}}{\sqrt{d_L(u) + d_R(u)}} \sim \mathcal{N}(0, \sigma_k^2)$
$\mathbf{x}_k + c\mathbf{1} \mapsto C_{k,u}$ $(\perp \mu_k)$
B 1 Stream 1
$C_{k,u} \in \mathbf{C}_{\mathrm{int}}$
B $l(u)$
$H_{k,l}, v_l, \Delta t_l$
Left Child $l(u)$
Left Gate ($g_{L,u}$):
$d_L(u) = (v_{l(u)} + \Delta t_{l(u)})\, e^{g_{L,u}}$
$g_{L,u} = 0 \implies d_L^{\mathrm{BM}} = v_l + \Delta t_l$
$d_{\mathrm{sum}}(u) = d_L + d_R$
Stream 2 — Convex State ($H_{k,u}$):
$w_L = \frac{d_R(u)}{d_{\mathrm{sum}}(u)}, \quad w_R = \frac{d_L(u)}{d_{\mathrm{sum}}(u)}$
$H_{k,u} = w_L H_{k,l(u)} + w_R H_{k,r(u)}$
$w_L + w_R \equiv 1 \implies \mathbf{x}_k + c\mathbf{1} \mapsto H_{k,u} + c$
B 1 Stream 2
$H_{k,u} \in \mathbf{H}_{\mathrm{nodes}}$
B $r(u)$
$H_{k,r}, v_r, \Delta t_r$
Right Child $r(u)$
Right Gate ($g_{R,u}$):
$d_R(u) = (v_{r(u)} + \Delta t_{r(u)})\, e^{g_{R,u}}$
$g_{R,u} = 0 \implies d_R^{\mathrm{BM}} = v_r + \Delta t_r$
Subtree Variance (v_u):
$v_u = \frac{d_L(u)\, d_R(u)}{d_L(u) + d_R(u)} = (d_L^{-1} + d_R^{-1})^{-1}$
$u = \rho = 294 \implies v_{294} = v_\rho = 0.2022$
1 1 Parent
$v_u \to p(u)$